

Fish assemblages in mangrove environments in South Sulawesi, Indonesia

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1 ABSTRACT

Mangroves are an essential biome for many tropical ecosystems and coastal communities. They exhibit high carbon sequestration, protect coastal areas from floods and extreme weather events, and foster biodiversity in their habitats. They have also been shown to increase fish stocks in surrounding fisheries. As the country with the most extensive mangrove habitats in the world, the Indonesian government has committed to restoring 600,000 hectares of mangroves throughout the country. One such place where this restoration effort is taking place is Pulau Tanakeke in South Sulawesi. Here, an NGO called Blue Forests has been conducting mangrove restoration since 2012. In the project, we use Unbaited Remote Underwater Video to assess how these replanting efforts are effecting local fish stocks and biodiversity. We compare three types of mangrove forests: natural (old-growth forests), EMR (replanted mangroves), and areas with destroyed mangroves. While we were unable to demonstrate a clear statistical relationship between our sampling sites in regards to fish diversity and abundance, this project serves as a baseline survey of the fish groups that live in these ecosystems in South Sulawesi. The benefits of mangroves for small scale fisheries is multi-dimensional and this research highlights the importance of environmental restoration for community engagement and ensuring food security.

2 INTRODUCTION

Mangroves are inter-tidal forests that are essential components to many tropical ecosystems. As the effects of climate change grow stronger worldwide, the need for carbon mitigation and protection against extreme weather are becoming more urgent. These biomes comprise about 14 % of marine carbon sequestration and may result in high gas emissions when these ecosystems are disturbed (Daniel M. Alongi 2012), while more established mangroves are more efficient in absorbing atmospheric carbon (Cameron, Hutley, and Friess 2019). Beyond their benefits of protecting against extreme weather events, mangroves are also key actors in maintaining the biodiversity of the ecosystems they inhabit. Mangroves have been reported to support up to 20% of the benthic biodiversity in their habitats (Carugati et al. 2018). They provide essential nutrients, temperature controls, and protection from predators for marine life (Blue Forests 2012). Further, Mangroves have been shown to increase fishery yields in their surrounding areas, therefore increasing fisher income (Aburto-Oropeza et al. 2008). The root systems of mangroves provide shelter and protection for juvenile fish, allowing them to grow and develop safely away from predators and also act as a buffer against strong currents and waves, creating calmer and more stable environments where fish can feed and reproduce (Daniel M. Alongi 2008). Areas with intact mangrove forests have been shown to support higher fish abundance

and diversity compared to areas without mangroves (Nagelkerken et al. 2008). Mangroves provide a rich food web, with leaf litter and detritus serving as a source of nutrients that fuel the basis of the food chain, supporting the growth and survival of various fish species (Daniel M. Alongi 2008). Furthermore, mangroves act as a buffer against coastal erosion and storm surges, safeguarding the habitats of both fish and fishermen (Nagelkerken et al. 2008). Mangroves offer a crucial line of defense against the impacts of climate change on fisheries. The dense root systems of mangroves stabilize shorelines and protect coastal areas from erosion caused by rising sea levels and extreme weather events (Daniel M. Alongi 2008). As the largest archipelago in the world, marine fisheries are an extremely important resource in Indonesia for food security and fisher income. Fisheries contribute to about 3% of the GDP in Indonesia, and over 80% of fishery catches are from small scale fisheries, however these systems are currently subject to over-exploitation, threatening the food security of those who are reliant on these fisheries (Ayunda, Sapota, and Pawelec 2018).

Despite all of their contributions to ecosystem health, mangrove environments are being threatened worldwide. Rising sea-levels has been shown to be a major contributor to mangrove loss (Gilman et al. 2008). Further, as extreme events are becoming more intense and more frequent, these could potentially threaten mangroves due to defoliation, soil erosion, or by altering the chemical makeup or temperature of soils (Gilman et al. 2008). Mangroves are also directly threatened by anthropogenic activity. Pollution, coastal development, and aquaculture development have also contributed to mangrove ecosystem loss (Adeel and Pomeroy 2002). Mangrove forests in the Western Tropical Pacific are the most diverse of these habitats globally (Ellison, Farnsworth, and Merkt 1999). Indonesia has the most extensive mangrove forests in the world (Kusmana 2011). As Indonesia is also the largest archipelagic country, mangroves' contribution to flood protection and extreme weather mitigation is vital to the health and safety of many of its inhabitants. However, due to timber production, aquaculture, and human development, the Ministry of Forestry reported in 2007 that around 69% of mangroves were in damaged condition in the country (Kusmana 2011). As a response to this habitat loss, the Indonesian government has committed to restoring 600,000 hectares of mangroves in the country in 2024, the most ambitious mangrove restoration project in the world.

Tanakeke Island in South Sulawesi, Takalar Regency, is one location where this restoration effort is taking place. Here, mangrove habitats are generally privately owned, and the main driver of mangrove loss has been timber production and shrimp aquaculture (Blue Forests 2012). 1,200 hectares of mangroves were converted into aquaculture ponds in the 1980s and 1990s, which is about 70% of its historical cover (Brown et al. 2014). Recently, there have been reports of increased flooding on this island which could be attributed to the loss of protection from mangroves (Blue Forests 2012). However, in response to mangrove restoration, the community has seen great success in both engagement and successful mangrove rehabilitation, where

hundreds of community members have participated since the 1990's (Blue Forests 2012; "Past and Present on Tanakeke Island" 2013). Further, small scale fishing is an essential part of peoples' livelihoods on Tanakeke. Locally caught seafood is the main source of protein on this island (Blue Forests 2012). Mangroves provide refuge for about 55% fish catch biomass in Indonesia (World Bank 2022). Fisher income has been shown to be negatively affected by mangrove habitat loss, and because of this, the financial gain of mangrove restoration is predicted to be more lucrative than any alternative land use such as aquaculture (Yamamoto 2023). For this reason, mangroves are an essential component to the health of Indonesia's fisheries and contribute significantly to the food security of those directly reliant on small scale fisheries as a key source of nutrition, including on Tanakeke Island (Blue Forests 2012). While these mangrove restoration efforts have resulted in increased flood prevention and ecotourism, little research has been done to assess the marine biodiversity and benefits to fish stocks. In this project, we aim to understand how this restoration is affecting local marine biodiversity and abundance of fish species on Tanakeke Island. We will use Unbaited Remote Underwater Video (URUV) methods to assess the marine biodiversity of mangrove habitats and compare them to the biodiversity of areas that have not undergone mangrove rehabilitation.

3 METHODS

3.1 Site description:

Pulau Tanakeke (Figure 1) is situated about 40 km southwest of Makassar in South Sulawesi, Takalar Regency. The island is a coral atoll covering about 3,930 hectares. About 392.25 hectares of mangrove has been restored (Cameron, Hutley, and Friess 2019). Access to the island presents a challenge as rough seas make sea crossings difficult during rainy seasons. The island comprises of six villages or desa: Mattirobaji, Lantang Peo, Tompotana, Maccinibaji, Balangdatu, and Rewataya. Historically, the island was populated with about 1,776 hectares of Mangroves, most of which has been destroyed for aquaculture.

3.2 URUVs

The study was conducted on Tanakeke Island in the Takalar Regency of South Sulawesi, Indonesia in the villages of Tompotanah and Lantangpeo (Fig PETA_AIS). In each village, mangroves were classified into three type: degraded, natural and EMR. EMR (Ecological Mangrove Restoration) refers to a low-cost, socially focused mangrove restoration method used by Blue Forests where local communities are involved in every step of the restoration process, and various mangrove replanting techniques are applied to the

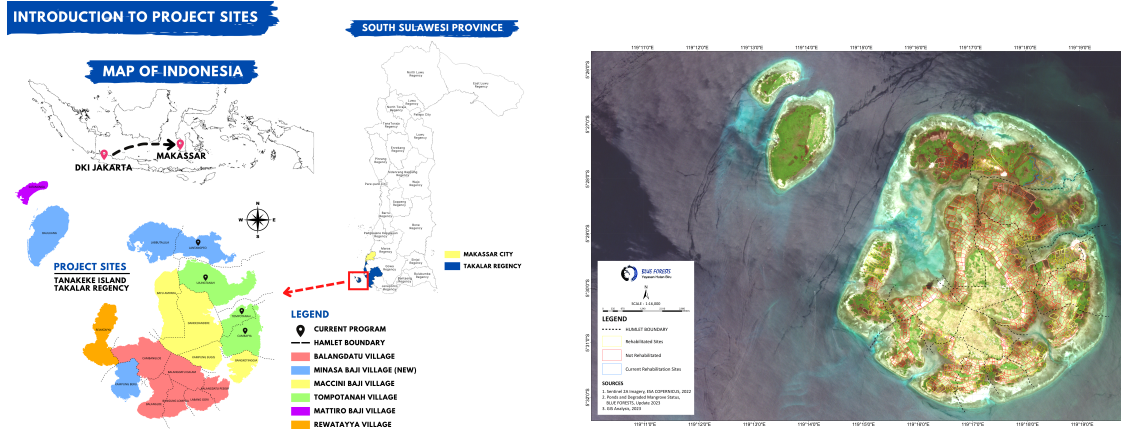


Figure 1: A map of Tanakeke Island with village (desa) names (left). The right image shows the mangrove restoration sites on Tanakeke Island by Blue forests. Both images are courtesy of Blue Forests.

restoration site (Brown et al. 2014). Three drop sites were selected in each mangrove type in each village based on mangrove type, water depth, and permission from land owners. Further, cameras were placed next to vegetation because this reduced the likelihood of boat traffic knocking over camera stands.

URUV methods followed standard BRUV methods outlined in (Langlois et al. 2020) but bait was not used. GoPro SJCAM SJ4000 Action Camera 4K30fps WiFi Cameras were installed on a 1 meter-tall PVC pipe partially buried in the sediment. Two villages were sampled and three cameras were installed in each village in each mangrove type, a total of 18 sites (PETA_AIS). The cameras were deployed horizontally to the ground with an additional 1 m PVC pipe extending parallel to the ground below the camera so that the length of that pipe was visible in the video. Tick marks spaced 10 cm apart were painted onto the pipe visible in the video. This allowed for a standardization of water visibility, as the number of ticks was counted for each video vile. Because this region experiences a wide tidal range, with mangrove forests being fully exposed to air at low tide, camera depths varied as they were deployed at high tide between the hours of 06:00 and 18:00 to ensure sufficient sunlight and that the camera stayed fully submerged during the duration of filming. Cameras faced vegetation in all sites. Site locations were chosen based on low-boat traffic and consent of the landowner. Date, time of day, tide height, water height at location, weather, and water temperature were recorded at each data collection.

3.3 Video Analysis

Camera calibrations were conducted by counting the number of tick marks visible on the extending PVC pipe. Videos were then cut to approximately one hour, starting from five minutes after the camera was set. Videos were watched manually on Windows Media Player and MaxN and T1st were counted and recorded

for each fish species found on the video. Fish were identified down the lowest possible taxonomic level and crabs were identified at the family-level. Gastropods were not counted as video-clarity was not sufficient to get accurate counts. A sub-sample of images were cross checked in order to avoid misidentified fish species, and species were identified using CITE DROPBOX BOOKS, BOOK ON SHELF. Further, we gathered input from members from both villages, Tompotanah and Lantangpeo, to collect locally-used names in the Makassar Language.

THIS WILL CHANGE BASED ON WHAT YOU END UP REPORTING After video data was analyzed and fish species were identified and counted, data was analyzed using RStudio software (Posit team 2024). At each village and mangrove type, total fish, total species of fish, Shannon’s Diversity Index and Simpson’s Diversity Index (Magurran 1988) was calculated. We then analyzed trends over time, as well as differences between the mangrove types with regards to fish diversity and assemblages. Finally, trophic scores were tabulated using information from FishBase (Boettiger, Temple Lang, and Wainwright 2012) to determine if different mangrover types were supporting different components of marine food chains.

4 RESULTS

Table 1: Animal groups identified, including local names used by villages in Tanakeke. Both members from Tompotanah (TMP) and Lantangpeo (LTP) were asked, so if answers differed, both were included in the local name.

* Fish group caught to be used as bait.

** Fish group caught for decorative purposes.

Group Name	ID Level	Common Name (Eng)	Common Name (Ind)	Pulau Tanakeke Local Name	Locally Caught
Acrochordus granulatus	Species	Marine File Snake		Care-care	No
Aronthron	Genus	Puffer Fish	Ikan Buntal	Buntal	Yes
Caranx	Genus	Trevally	Ikan Kuwe	Juku kebo	Yes
Chaetodon	Genus	Butterfly Fish	Ikan Kepe-kepe	Ikan karang	Yes**
Fibramia lateralis	Species	Humpback Cardinal		Besing-besing	Yes*
Gerres oyena	Species	Blacktip Silverbiddy	Ikan Kapas-kapas	Birangkasa	Yes
Hyporhamphus	Genus	Halfbeak	Ikan julung-julung	Ikan julung-julung	No
Lutjanus argentimaculatus	Species	Mangrove Red Snapper	Ikan Lencam	Katamba (LTP) Sikuda (TMP)	Yes
Lutjanus ehrenbergii	Species	Blackspot Snapper	Kakap Garis	Ganrang eja	Yes
Monodactylus argenteus	Species	Moonyfishes or fingerfishes	Ikan Bebulan	Ikan Tapi-tapi	No
Monodactylus argenteus	Species	Diamondfish		Ikan Tapi-tapi	No
Mugilidae	Family	Mullet	Ikan Belanak	balanak	Yes
Omobranchus	Genus	Blenny	Jabingan	Ikan toto	No
Ostorhinchus ocellicaudus	Species	Tailspot cardinalfish		Tontong	No
Scatophagus argus	Species	Spotted Scat	Ikan Kiper	Titam/Titang	Yes
Siganus	Genus	Gold Spotted Rabbitfish	Ikan Baronang	Baronang	Yes
Sphaeramia orbicularis	Species	Orbicular Cardinalfish		Besing-besing	Yes
Sphyraena barracuda	Species	Great Barracuda	Barakuda	Berang belang (TMP)	Yes
Terapon jarbua	Species	Crescent Grunter	Kerong Kerong	Kerong-kerong	No
Toxotes jaculatrix	Species	Banded Archerfish	Ikan Sumpit	Berang belang (TMP) Gamasi (LTP)	No

In total, we identified 41 different animal groups, with 24 fish groups, 1 species of snake, 2 genres of crab, and various species of shrimp (which were grouped together due to difficulty of identification). Of these, we were able to identify 13 groups to the species level, 6 genres, and 1 groups to the family level. After groups were identified down to their lowest taxonomic level, we asked fishers of both villages for the local name used for these fish (Table 1).

Viewing the data in a time series, both diversity indices showed a low point in June, whereas the total MaxN for that month was among the highest (Figure 2). This could indicate that June is a month with high fish counts, but may be dominated by only a few species.

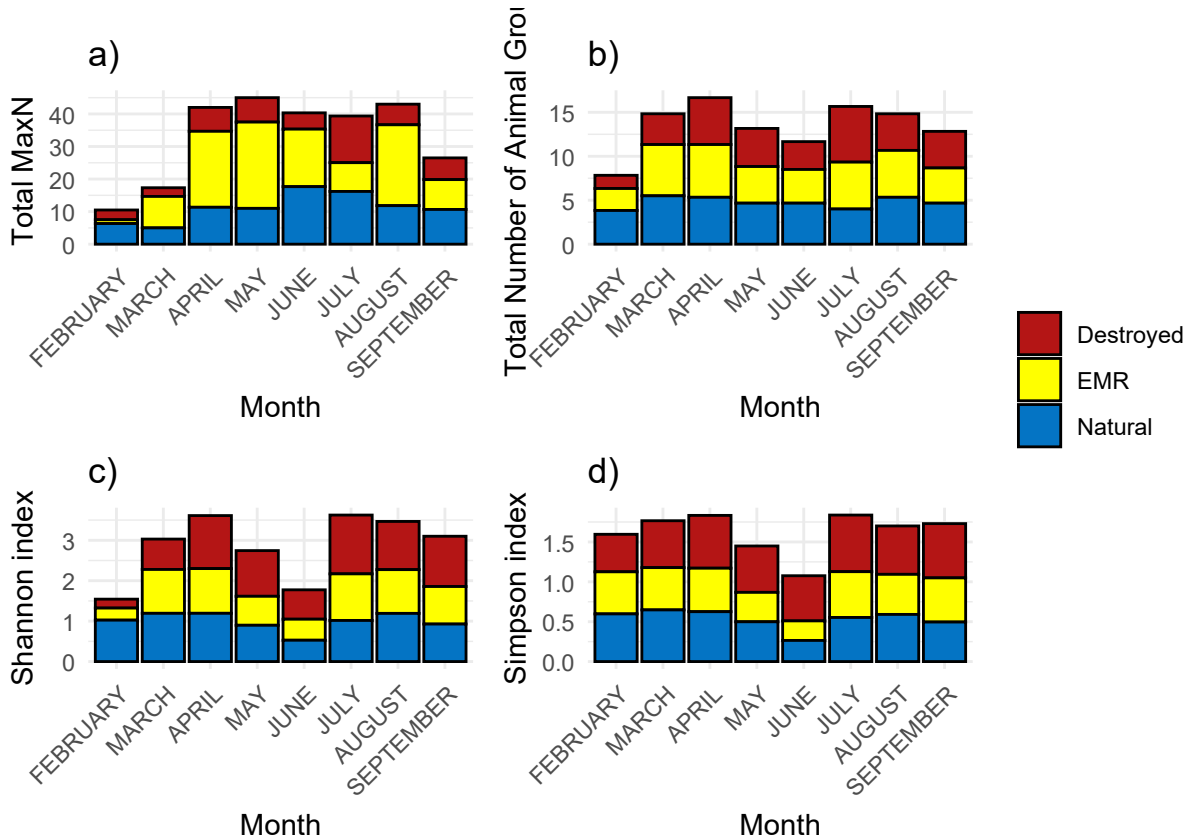


Figure 2: Monthly counts of a) total MaxN among all animal groups found. b) Number of animal groups found, c) Shannon's Diversity Index, d) Simpson's Diversity Index

Next, as mangrove replanting efforts on Tanakeke could be benefiting local fish communities, we calculated the Shannon Diversity Index, Simpson Diversity Index, overall MaxN, and total number of fish species found in each site. However, our results did not suggest a significant relationship between site type and fish abundance and diversity (Figure 3). We also tested the proportion of occurrences of each species in each site type to indicate where species tended to prefer to inhabit. Here, a stronger relationship was uncovered where more species actually seemed to prefer the EMR sites over the destroyed or natural sites (Figure 4).

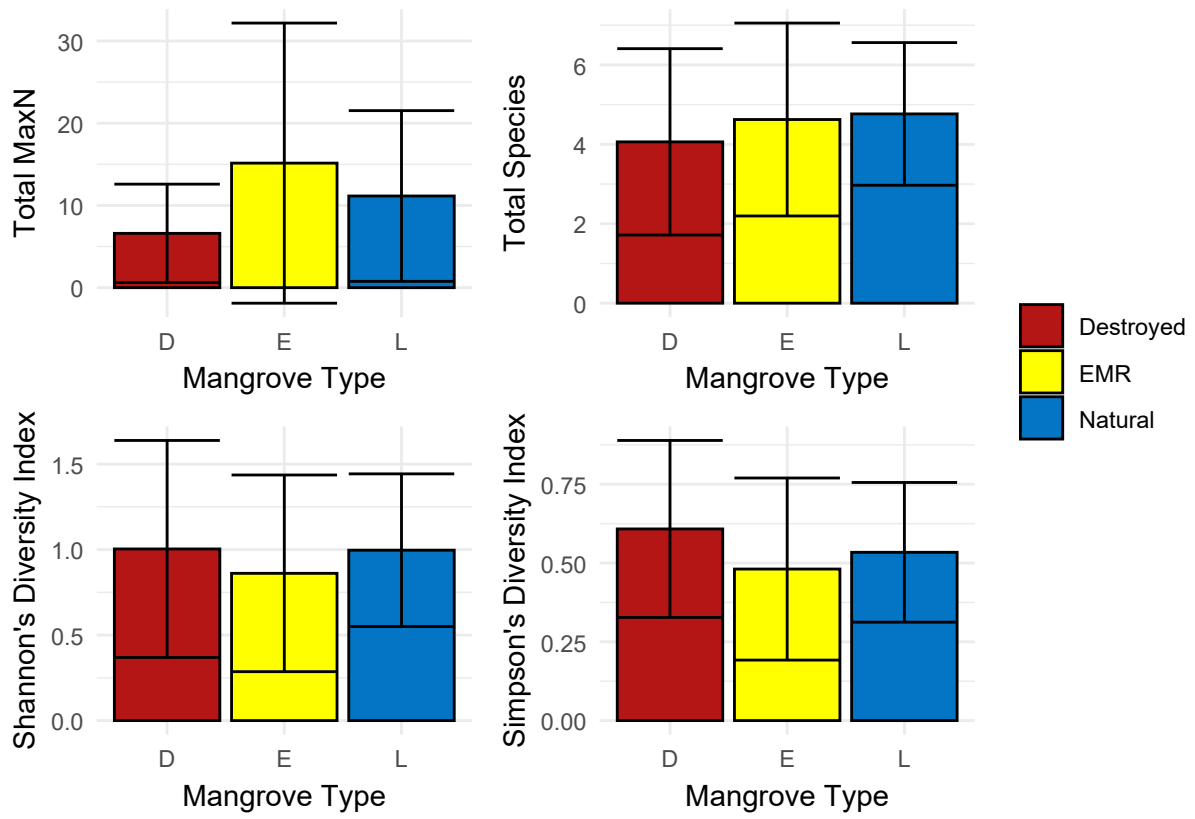


Figure 3: Mean and standard deviation of a) total MaxN among all animal groups found. b) Number of animal groups found, c) Shannon's Diversity Index, d) Simpson's Diversity Index in each mangrove type

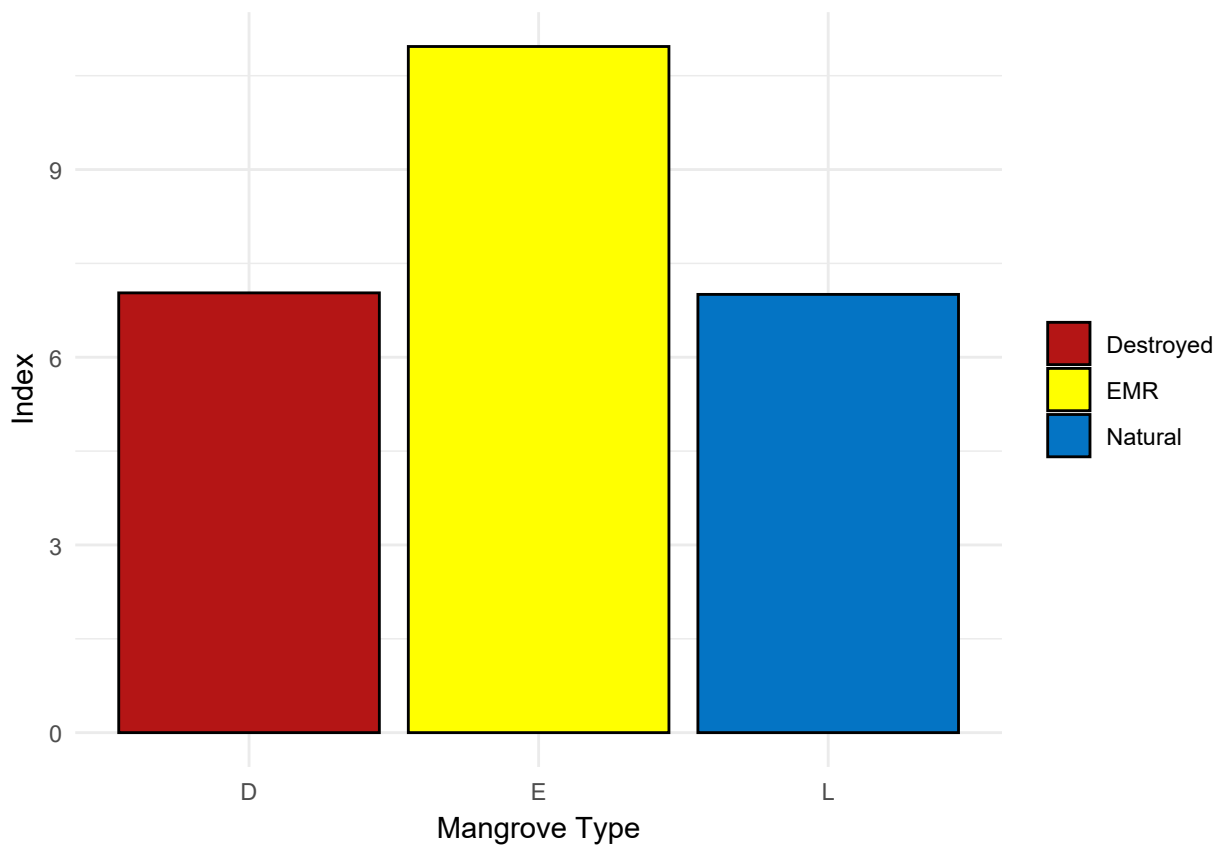


Figure 4: Habitat preference of each fish group. This is the proportion of each group that were found in each site type.

After identifying fish groups, we used the trophic score reported by FishBase for that species or genus to calculate the trophic level of fish that were found in each site(Figure 5). Here, we found that the EMR sites had a higher proportion of fish of higher trophic levels when compared to the destroyed or natural sites, which were more balanced across the foodweb.

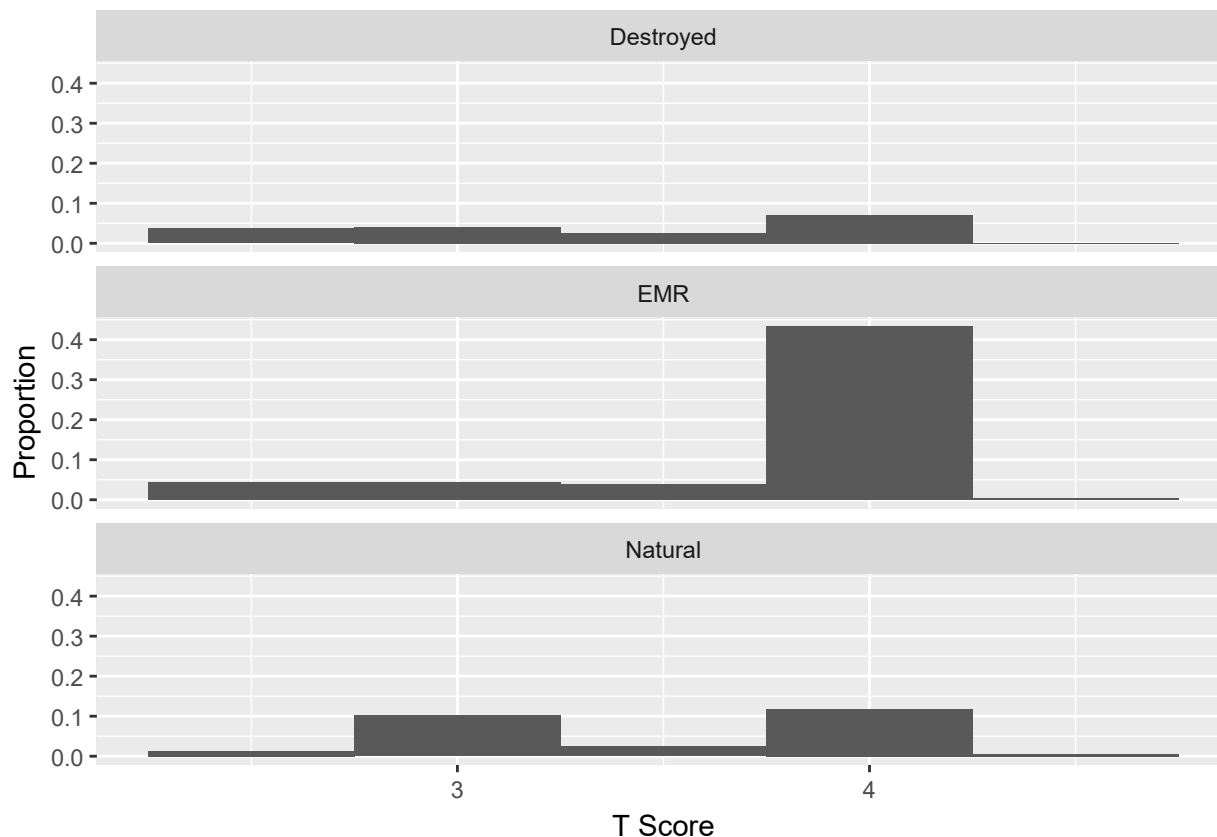


Figure 5: Again, I acknowledge that this is not the best way to present this information but I was looking into the trophic level supported by each site type

5 DISCUSSION

Although our findings didn't demonstrate a clear relationship between habitat type and fish abundance or biodiversity, we were able to characterize the fish species that inhabit these regions, as well as which species were of interest to the local community. Further, as June was a month with high fishcounts yet low diversity, this could indicate a possible breeding season as most of these fish were either cardinal fish, or unidentifiable juveniles. SOMETHING ABOUT MANGROVES SUPPORTING JUVENILES, SOMETHING ABOUT UNDERSTANDING JUVENIL DYNAMICS HELPS UNDERSTAND FISHERIES BETTER AS A WHOLE. However, identifying juveniles from video analysis alone is difficult and certainly a challenge in this study.

Stereo-BRUVS or stereo-URUVS, which are video techniques that set two, inward facing cameras to get size estimations of fish, offer one potential opportunity to better study these assemblages (Piggott et al. 2020) MAYBE REPHRASE AND SAY THIS IS ONE OF THE BETTER METHODS.

EMR sites showed slightly different dynamics from both the destroyed and natural sites, indicating these areas may be in a temporary imbalance, as most fish species actually preferred these sites, and these areas had a higher proportion of fish with higher trophic scores. As mangroves regrow and the local ecosystem transitions from open water with sandy and rocky benthos to areas with vegetation and SILTY WATER BOTTOMS, the local fauna may be in transition as well. This is especially important on Tanakeke, where substrate contents, tidal patterns, and natural lack of mangrove diversity actually make restoration particularly challenging CITATION SPECIFIC TO TANAKEKE (Cameron, Hutley, and Friess 2019; Kusumaningtyas et al. 2022). This can influence not only the efficacy of restoration efforts, but also the ecosystem function of mangroves as they regrow (Cameron et al. 2019).

Although we were unable to show statistical significance in the difference in fish biodiversity among different mangrove sites on Tanakeke, the benefits of mangroves to local communities and local biodiversity is a WELL SUPPORTED SOMETHING. HERE TALK ABOUT BROWN ET AL AND THEIR APPROACH. ALSO WOMANGROVE ARTICLES It is difficult to fully quantify the benefits of mangroves to fisheries. Fisheries themselves are usually classified by the catch, not the ecosystems they rely on so understanding the full benefits of mangroves on fish is difficult to encompass at large scale due to the multi-dimensionality of benefits and the lack of coherent data (Seary et al. 2021). Because of this, mangroves are often undervalued and converted to other land uses CITATION 52 IN searyDefiningMangrovefisheriesTypology2021.

As many studies have found that the majority of tropical fish species benefit from mangroves at some point in their life cycle, we think that our lack of statistical relationships between sites could be due to sampling design. Sites were selected based on habitat type, but the actual placement of the URUVs was INFLUENCED by other factors such as explicit permission from the landowner and the sites needed to be accessible from the boat and have clear video coverage of the area (i.e. extremely dense mangrove forests that couldn't be accessed or too dark for the video footage). Further, even in destroyed sites, cameras had to be placed near vegetation in order to avoid boat and foot traffic that could influence the fish behavior and location while filming. Further, due to funding and resource availability, we were only able to collect data for 8 months, whereas a full year of data could paint a more robust picture of fish behavior and preference.

This study provides a knowledge base for the fish diversity in mangrove forests of Tanakeke Island, Indonesia. Since mangrove restoration is a critical undertaking, especially in Indonesia, this study serves as baseline understanding of fishery and mangrove function on Tanakeke. Long-term monitoring of restored mangroves

has helped quantify the long-term economic benefits and environmental services of mangroves (Wirabuwana et al. 2025). Conducting long-term surveys of fish assemblages in this ear could help better quantify the lasting impacts of restoration efforts. MORE LONG TERM STUFF

6 ACKNOWLEDGEMENTS

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References

- Aburto-Oropeza, Octavio, Exequiel Ezcurra, Gustavo Danemann, Víctor Valdez, Jason Murray, and Enric Sala. 2008. “Mangroves in the Gulf of California Increase Fishery Yields.” *Proceedings of the National Academy of Sciences* 105 (30): 10456–59. <https://doi.org/10.1073/pnas.0804601105>.
- Adeel, Zafar, and Robert Pomeroy. 2002. “Assessment and Management of Mangrove Ecosystems in Developing Countries.” *Trees* 16 (2-3): 235–38. <https://doi.org/10.1007/s00468-002-0168-4>.
- Alongi, Daniel M. 2012. “Carbon Sequestration in Mangrove Forests.” *Carbon Management* 3 (3): 313–22. <https://doi.org/10.4155/cmt.12.20>.
- Alongi, Daniel M. 2008. “Mangrove Forests: Resilience, Protection from Tsunamis, and Responses to Global Climate Change.” *Estuarine, Coastal and Shelf Science* 76 (1): 1–13. <https://doi.org/10.1016/j.ecss.2007.08.024>.
- Ayunda, Nisa, Mariusz R. Sapota, and Anna Pawelec. 2018. “The Impact of Small-Scale Fisheries Activities Toward Fisheries Sustainability in Indonesia.” In *Interdisciplinary Approaches for Sustainable Development Goals*, edited by Tymon Zielinski, Iwona Sagan, and Waldemar Surosz, 147–67. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-319-71788-3_11.
- Blue Forests. 2012. “Adaptive Collaborative Management Plan for Building Mangrove Resilience in Tanakeke Island.”
- Boettiger, Carl, Duncan Temple Lang, and Peter Wainwright. 2012. “Rfishbase: Exploring, Manipulating and Visualizing FishBase Data from r.” *Journal of Fish Biology*, November. <https://doi.org/10.1111/j.1095-8649.2012.03464.x>.
- Brown, Ben, Gaëll Mainguy, Ratna Fadillah, Yusran Nurdin, Iona Soulsby, and Rio Ahmad. 2014. “CASE STUDY: Community Based Ecological Mangrove Rehabilitation (CBEMR) in Indonesia.”
- Cameron, Clint, Lindsay B. Hutley, and Daniel A. Friess. 2019. “Estimating the Full Greenhouse Gas Emissions Offset Potential and Profile Between Rehabilitating and Established Mangroves.” *Science of The Total Environment* 665 (May): 419–31. <https://doi.org/10.1016/j.scitotenv.2019.02.104>.
- Cameron, Clint, Lindsay B. Hutley, Daniel A. Friess, and Niels C. Munksgaard. 2019. “Hydroperiod, Soil Moisture and Bioturbation Are Critical Drivers of Greenhouse Gas Fluxes and Vary as a Function of Landuse Change in Mangroves of Sulawesi, Indonesia.” *Science of The Total Environment* 654 (March): 365–77. <https://doi.org/10.1016/j.scitotenv.2018.11.092>.
- Carugati, Laura, Beatrice Gatto, Eugenio Rastelli, Marco Lo Martire, Caterina Coral, Silvestro Greco, and Roberto Danovaro. 2018. “Impact of Mangrove Forests Degradation on Biodiversity and Ecosystem Functioning.” *Scientific Reports* 8 (1): 13298. <https://doi.org/10.1038/s41598-018-31683-0>.

- Ellison, Aaron M., Elizabeth J. Farnsworth, and Rachel E. Merkt. 1999. "Origins of Mangrove Ecosystems and the Mangrove Biodiversity Anomaly: Mangrove Biodiversity Anomaly." *Global Ecology and Biogeography* 8 (2): 95–115. <https://doi.org/10.1046/j.1466-822X.1999.00126.x>.
- Gilman, Eric L., Joanna Ellison, Norman C. Duke, and Colin Field. 2008. "Threats to Mangroves from Climate Change and Adaptation Options: A Review." *Aquatic Botany* 89 (2): 237–50. <https://doi.org/10.1016/j.aquabot.2007.12.009>.
- Kusmana, Cecep. 2011. "Management of Mangrove Ecosystem in Indonesia."
- Kusumaningtyas, Mariska Astrid, Terry L. Kepel, Tubagus Solihuddin, Ali Arman Lubis, Aditya Dwi P. Putra, Untung Sugiharto, Restu N. A. Ati, et al. 2022. "Carbon Sequestration Potential in the Rehabilitated Mangroves in Indonesia." *Ecological Research* 37 (1): 80–91. <https://doi.org/10.1111/1440-1703.12279>.
- Langlois, Tim, Jordan Goetze, Todd Bond, Jacquomo Monk, Rene A. Abesamis, Jacob Asher, Neville Barrett, et al. 2020. "A Field and Video Annotation Guide for Baited Remote Underwater Stereo-Video Surveys of Demersal Fish Assemblages." Edited by Edward Codling. *Methods in Ecology and Evolution* 11 (11): 1401–9. <https://doi.org/10.1111/2041-210X.13470>.
- Magurran, Anne E. 1988. "Diversity Indices and Species Abundance Models." In *Ecological Diversity and Its Measurement*, 7–45. Dordrecht: Springer Netherlands. https://doi.org/10.1007/978-94-015-7358-0_2.
- Nagelkerken, I., S. J. M. Blaber, S. Bouillon, P. Green, M. Haywood, L. G. Kirton, J.-O. Meynecke, et al. 2008. "The Habitat Function of Mangroves for Terrestrial and Marine Fauna: A Review." *Aquatic Botany* 89 (2): 155–85. <https://doi.org/10.1016/j.aquabot.2007.12.007>.
- "Past and Present on Tanakeke Island." 2013. *Blue Forests*.
- Piggott, Camilla V. H., Martial Depczynski, Monica Gagliano, and Tim J. Langlois. 2020. "Remote Video Methods for Studying Juvenile Fish Populations in Challenging Environments." *Journal of Experimental Marine Biology and Ecology* 532 (November): 151454. <https://doi.org/10.1016/j.jembe.2020.151454>.
- Posit team. 2024. *RStudio: Integrated Development Environment for r*. Boston, MA: Posit Software, PBC. <http://www.posit.co/>.
- Seary, Rachel, Tom Spencer, Mike Bithell, Chris McOwen, and Yoshitaka Ota. 2021. "Defining Mangrove-Fisheries: A Typology from the Perancak Estuary, Bali, Indonesia." Edited by Ismael Aaron Kimirei. *PLOS ONE* 16 (4): e0249173. <https://doi.org/10.1371/journal.pone.0249173>.
- Wirabuana, Pandu Y. A. P., Liliana Baskorowati, Bayu Pamungkas, Budi Mulyana, Josie South, Hery Purnobasuki, Sapto Andriyono, and Veryl Hasan. 2025. "Mangroves, Fauna Compositions and Carbon Sequestration After Ten Years Restoration on Flores Island, Indonesia." *Scientific Reports* 15 (1): 4866. <https://doi.org/10.1038/s41598-025-87307-x>.
- World Bank. 2022. "New Project Will Support Large-Scale Mangrove Conservation and Restoration in

Indonesia.” Text/{HTML}. *The World Bank*. <https://www.worldbank.org/en/news/press-release/2022/06/07/new-project-will-support-large-scale-mangrove-conservation-and-restoration-in-indonesia>.

Yamamoto, Yuki. 2023. “Living Under Ecosystem Degradation: Evidence from the Mangrove–Fishery Linkage in Indonesia.” *Journal of Environmental Economics and Management* 118 (March): 102788. <https://doi.org/10.1016/j.jeem.2023.102788>.